Design Proposal for Mach-Zehnder Interferometers

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I. INTRODUCTION

A Mach-Zehnder interferometer (MZI) is a device where a beam of coherent light is first split into two different beams, and then recombined, and the phase difference between the two paths taken by the two beams result in constructive or destructive interference. Based on this effect, one can route light to different channels by controlling the phase shift. Mach-Zehnder interferometers have been widely used for different applications. This device can be implemented in an integrated photonic platform using two y-splitters (or bidirectional couplers) and waveguides. In this document, the design of such a device is proposed and initial simulation results are provided.

II. THEORY

In a Mach-Zehnder interferometer, a component is needed to split the light into two paths, and then another component is needed to recombine. In free-space optics, this is done using beam splitters. For our integrated photonics case, a y-splitter is chosen for this purpose. Assuming a 50/50 splitting ration, and also assuming the two beam paths travel through identical media (therefore with the same propagation constant β), the output intensity can be expressed as

$$I_o = \frac{I_i}{4}[1 + cos(\beta \Delta L)], \tag{1}$$

where I_o is the intensity of the output, I_i is the intensity of the input and ΔL is the difference in the length of the path.

III. MODELING AND SIMULATION

In this section, simulation results for the waveguide used and the interferometer circuit.

A. Waveguide in Lumerical MODE

For the design, a waveguide width and height of 500 nm and 220 nm are chosen respectively. The first quasi-TE and quasi-TM mode profiles are shown in Fig. 1. The effective mode indices as a function of wavelength for the two modes are shown in Fig. 2, and the

group incides are shown in Fig. 3. The waveguide compact model, obtained by fitting a polynomial to the data in Fig. 2, for the quasi-TE mode is

$$n_{eff} = 2.44 - 1.13(\lambda - \lambda_0) - 0.04(\lambda - \lambda_0), \tag{2}$$

and that for the quasi-TM mode is

$$n_{eff} = 1.77 - 1.27(\lambda - \lambda_0) + 1.82(\lambda - \lambda_0), \tag{3}$$

where $\lambda_0 = 1550$ nm.

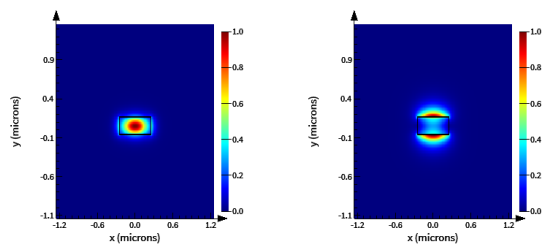

FIG. 1. Electric field intensity of quasi-TE mode (left) and quasi-TM mode (right) in the waveguide

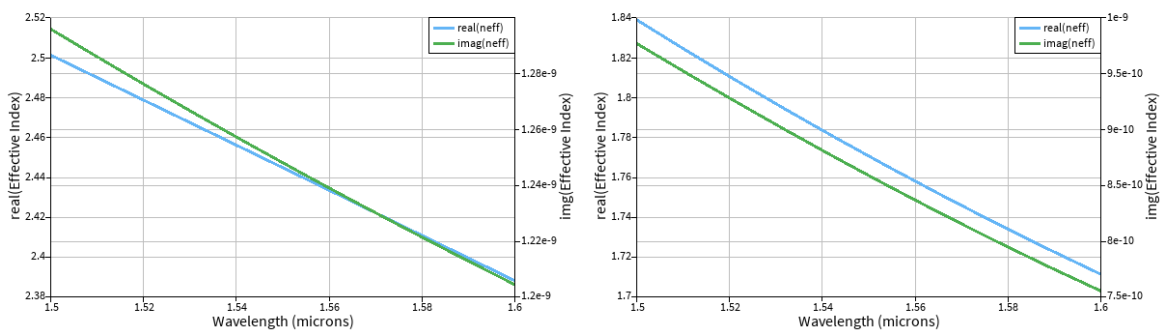

FIG. 2. The effective mode indices of the quasi-TE (top) and the quasi-TM (bottom) mode as a function of wavelength.

B. Mach-Zehnder Interferometer in Lumerical INTERCONNECT

The polarization for the circuit is determined by the grating coupler used. For the proposed circuits, the quasi-TE polarization is chosen. The circuit for the Mach-Zehnder interferometer is presented in Fig. 7. A bidirectional coupler is used for the design to visualize

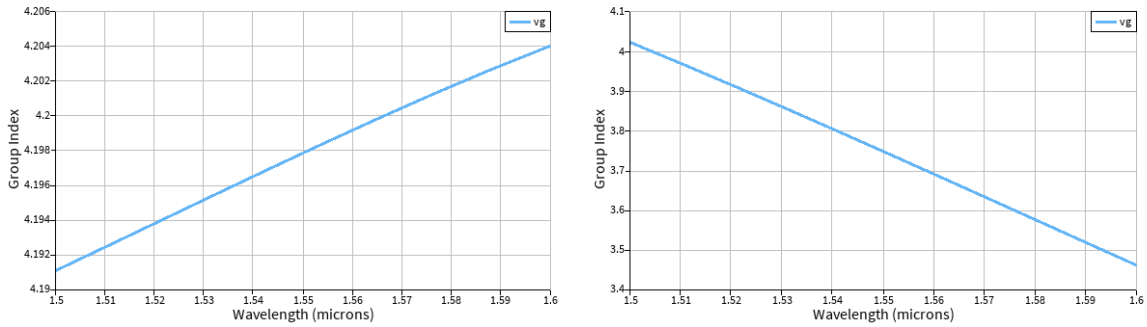

FIG. 3. The group indices of the quasi-TE (top) and the quasi-TM (bottom) mode as a function of wavelength.

both output modes from the interferometer. Fig. 5 presents the gain of the grating couplers, the y-splitter and the bidirectional coupler used. The gain of the full circuit with varying ΔL is shown in Fig. 6. ΔL is varied from 0 µm to 200 µm. The free-spectral range (FSR) of each circuit is tabulated in Table. I.

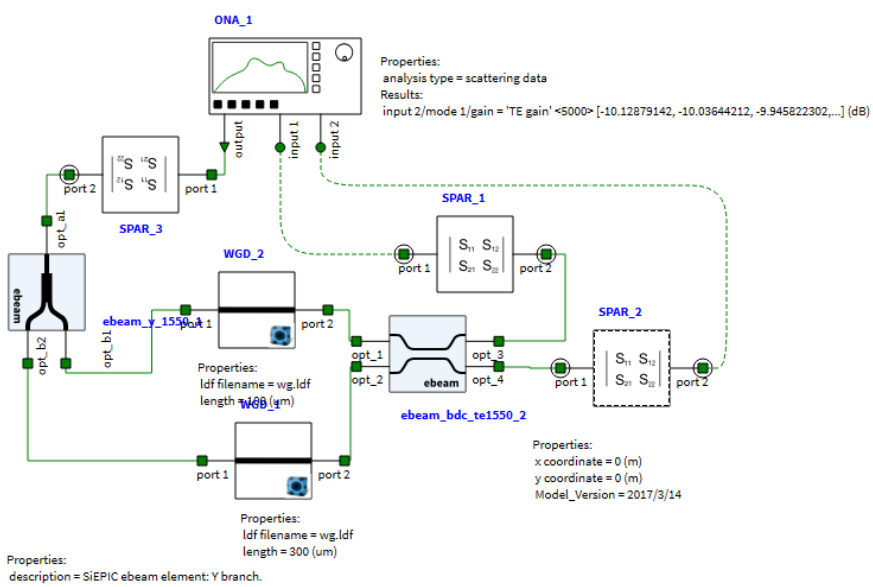

FIG. 4. Circuit diagram

C. Group Index Derivation

Here, we derive the relationship between the free-spectral range (FSR), which can be obtained from either the simulation results in Fig. 6 or the experimental data.

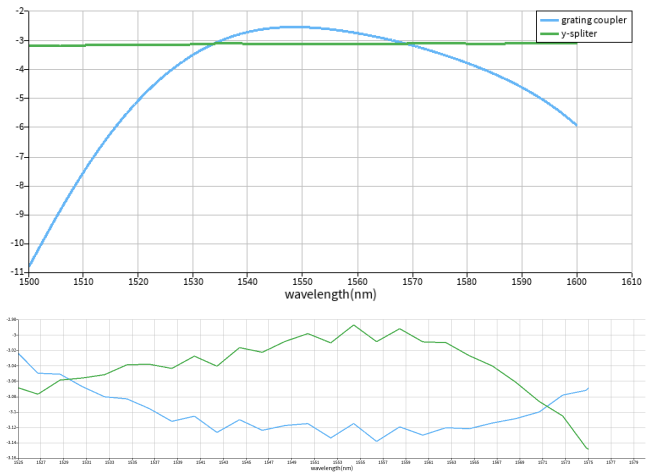

FIG. 5. Gain of the grating coupler (blue) and the y-splitter (green) are shown in the top figure. The bottom figure shows the gain of the bidirectional coupler.

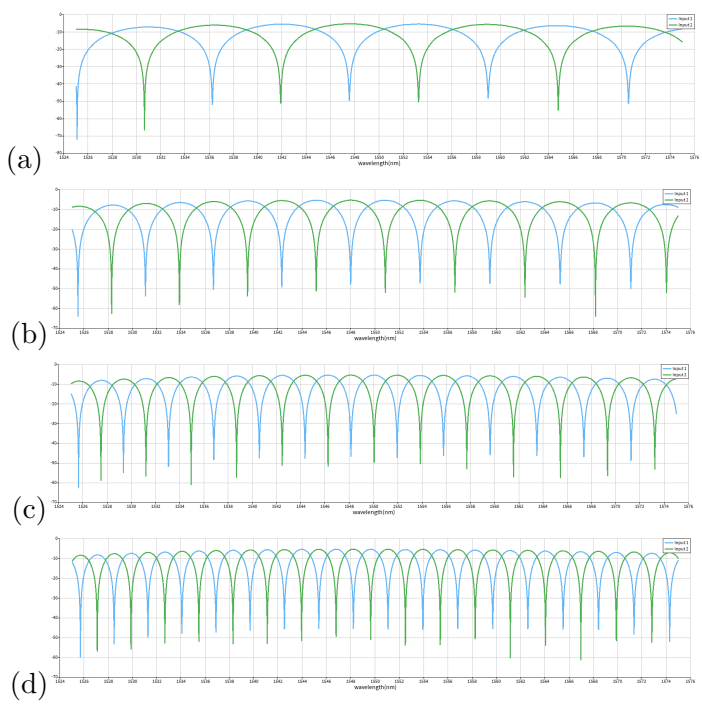

FIG. 6. Gain of the MZI with ΔL set to (a) 50 µm, (b) 100 µm, (c) 150 µm and (d) 200 µm.

The FSR is defined as the spacing between adjacent peaks in the interferometer spectrum

$$\mathrm{FSR} = \lambda_{m+1} - \lambda_m. \tag{4}$$

The phase difference between adjacent peaks is 2π, therefore from (1), we have

$$(\beta_m - \beta_{m+1})\Delta L = \Delta\beta = 2\pi. \tag{5}$$

ΔL µm	FSR (nm)
50	11.37
100	5.48
150	3.86
200	2.84

TABLE I. Free-spectral range for the different ΔL values.

Approximating a linear variation in *beta* with wavelength λ, i.e. $\Delta\beta \approx -\frac{d\beta}{d\lambda}\text{FSR}$, and using $\beta = 2\pi n/\lambda$ we get to

$$FSR = \frac{\lambda^2}{\Delta L n_g}. \tag{6}$$

Figure 7 shows the comparison of the FSR obtained from the simulations and from (6), and there is very good agreement.

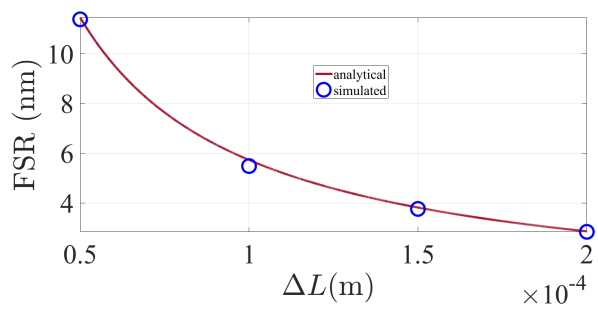

FIG. 7. Circuit diagram

IV. CONCLUSION

MZI interferometer circuits with TE polarization and four different path-length differences have been proposed. The compact model for the waveguides to be used have been extracted from Lumerical MODE simulations, and the circuits have been modeled in Lumerical INTERCONNECT. The FSR obtained from the simulations show good agreement with expected values.